

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4	9	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

~~LSE~~ Least square error = $L(\theta_1, \theta_2) = (\theta_1 + \theta_2 - y_1)^2 + (\theta_1 - 2\theta_2 - y_2)^2 + (2\theta_1 - \theta_2 - y_3)^2$

~~To find the LSE of θ_1, θ_2 we have to~~
 ~~$\frac{\partial L}{\partial \theta_1} = 0$~~
 ~~$\frac{\partial L}{\partial \theta_2} = 0$~~

Let $\hat{\theta}_1, \hat{\theta}_2$ be the LSEs of θ_1, θ_2 respectively.

~~The~~ $L(\hat{\theta}_1, \hat{\theta}_2)$ is the minimum possible LSE.

~~$\frac{\partial L}{\partial \theta_1}(\hat{\theta}_1, \hat{\theta}_2) = 0$~~
 ~~$\frac{\partial L}{\partial \theta_2}(\hat{\theta}_1, \hat{\theta}_2) = 0$~~

~~$\frac{\partial L}{\partial \theta_1}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\partial L}{\partial \theta_2}(\hat{\theta}_1, \hat{\theta}_2) = 0$~~

$\Rightarrow 2(\hat{\theta}_1 + \hat{\theta}_2 - y_1) + 2(\hat{\theta}_1 - 2\hat{\theta}_2 - y_2) + 2(2\hat{\theta}_1 - \hat{\theta}_2 - y_3) = 0$

$\Rightarrow 12\hat{\theta}_1 - 6\hat{\theta}_2 + 2y_1 - 2y_2 - 4y_3 = 0$

$\Rightarrow 12\hat{\theta}_1 - 6\hat{\theta}_2 = 2(y_1 + y_2 + 2y_3)$

$\Rightarrow \hat{\theta}_2 = 2\hat{\theta}_1 - \frac{y_1 + y_2 + 2y_3}{3}$

$$\frac{\partial L}{\partial \theta_2}(\hat{\theta}_1, \hat{\theta}_2) = 0$$

$$\Rightarrow 2(\hat{\theta}_1 + \hat{\theta}_2 - y_1) + 2(-2)(\theta_1 - 2\theta_2 - y_2) + 2(-1)(2\theta_1 - \theta_2 + y_3) = 0$$

$$\Rightarrow -6\hat{\theta}_1 + 12\hat{\theta}_2 = 2y_1 - 4y_2 - 2y_3$$

$$\Rightarrow -\hat{\theta}_1 + 2\left(2\hat{\theta}_1 - \frac{y_1 + y_2 + 2y_3}{3}\right) = \frac{y_1 - 2y_2 - y_3}{3}$$

$$\Rightarrow 3\hat{\theta}_1 = \frac{y_1 - 2y_2 - y_3}{3} + \frac{2y_1 + 2y_2 + 4y_3}{3} = y_1 + y_3$$

$$\Rightarrow \boxed{\hat{\theta}_1 = \frac{y_1 + y_3}{3}}$$

$$\hat{\theta}_2 = \frac{2y_1 + 2y_3}{3} - \frac{y_1 + y_2 + 2y_3}{3} = \boxed{\frac{y_1 - y_2 + y_3}{3}}$$

$$\text{LSE of } \theta_1 = \hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\text{L of } \theta_2 = \hat{\theta}_2 = \frac{y_1 - y_2 + y_3}{3}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) given uniform distribution $U(0,1) \Rightarrow \text{mean} \Rightarrow E(U) = \frac{1}{2}$ & $\text{Var}(U) = \frac{1}{12}$
~~Cov(X_1, X_2) = 0~~ $\text{Var}(X_1) = \text{Var}(X_2) = \frac{1}{12}$
 So,

$$\begin{aligned}\text{Var}(X_3) &= \text{Cov}(U_1 + U_2, U_1 + U_2) \\ &= \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2) \\ U_1, U_2 \text{ are independent, so } \text{Cov}(U_1, U_2) &= 0\end{aligned}$$

$$\Rightarrow \text{Var}(X_3) = \frac{1}{12} \times 2 = \frac{1}{6}$$

$$\begin{aligned}\text{Cov}(X_1, X_2) &= 0 \quad (U_1, U_2 \text{ are independent}) \\ \Rightarrow \boxed{\rho_{X_1, X_2} = 0} \\ \text{Cov}(X_1, X_3) &= \text{Cov}(X_1, X_1 + X_2) = \text{Cov}(X_1, X_1) + \text{Cov}(X_1, X_2) \\ &= \frac{1}{12} + 0 \\ &= \frac{1}{12}\end{aligned}$$

$$\rho_{X_1, X_3} = \frac{\text{Cov}(X_1, X_3)}{\sqrt{\text{Var}(X_1)}\sqrt{\text{Var}(X_3)}} = \frac{1/12}{\sqrt{1/12} \times \sqrt{1/6}}$$

$$\Rightarrow \boxed{\rho_{X_1, X_3} = \frac{1}{\sqrt{2}}}$$

$$\begin{aligned}\text{Cov}(X_2, X_3) &= \text{Cov}(X_2, X_1 + X_2) = \text{Cov}(X_2, X_1) + \text{Cov}(X_2, X_2) \\ &= 0 + \frac{1}{12} \\ &= \frac{1}{12}\end{aligned}$$

$$\boxed{\rho_{X_2, X_3} = \frac{\text{Cov}(X_2, X_3)}{\sqrt{\text{Var}(X_2)}\sqrt{\text{Var}(X_3)}} = \frac{1}{\sqrt{2}}}$$

03 $\therefore \rho(\mathbf{X}) = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$ (Correlation matrix of $\mathbf{X}$)

(b). We have to find eigenvalues of $p(x)$.

Let λ be an eigenvalue of $p(x)$.

$$\Rightarrow p(x)v = \lambda v \text{ for some vector } v \in \mathbb{R}^3$$

$$\Rightarrow |p(x) - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{1}{2} \right] + \frac{1}{\sqrt{2}} \left[-\frac{1}{\sqrt{2}} (1-\lambda) \right] = 0$$

$$\Rightarrow (1-\lambda) \left[\lambda + \lambda^2 - 2\lambda - \frac{1}{2} \right] = 0$$

$$\Rightarrow (1-\lambda)\lambda(\lambda-2) = 0.$$

$\Rightarrow$ eigenvalues of $p(x) \Rightarrow \{1, 0, 2\}$.

$\lambda=2$ gives the principal component y_1 .

Let v_2 be the corresponding eigenvector of $\lambda=2$.

$$y_1 = v_2' X$$

~~Var(y_1) = 2~~ (the eigenvalue)

~~Let $v_2' = [a \ b \ c]$~~

$$\Rightarrow p(x)v_2 = 2v_2 \Rightarrow \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 2 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\Rightarrow a + c/\sqrt{2} = 2a \Rightarrow a = c/\sqrt{2}$$

$$b + c/\sqrt{2} = 2b \Rightarrow b = c/\sqrt{2}$$

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix}$$

Var(y_1) = $v_2' \text{Var}(X) v_2$

$$= \frac{1}{8} + \frac{1}{6\sqrt{2}}$$

$$y_1 = \left(\frac{x_1}{2} + \frac{x_2}{2} + \frac{x_3}{\sqrt{2}} \right)$$

~~Var(y_1) = 2~~ (corresponding eigenvalue)

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a). (i). $d(2,1) = 1$ is the lowest distance observed, so we can cluster them to (12).

Now, distance matrix is:

$$\begin{matrix} & (12) & 3 & 4 \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 3.5 \end{bmatrix} \end{matrix}$$

$$d((12), 3) = \max(d(1,3), d(2,3)) = \{11, 2\} = 11$$

{using complete linkage hierarchical procedure}

The least distance now is between 3, 4 $\Rightarrow d(3,4) = 3.5$

$$d((12), 4) = \max(d(1,4), d(2,4)) = \{5, 3\} = 5$$

So, we can cluster them to (3,4)

Now, distance matrix is:

$$\begin{matrix} & (12) & (34) \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 \\ 11 \end{bmatrix} & \begin{bmatrix} 0 \end{bmatrix} \end{matrix}$$

$$d((12), (34)) = \max(d((12), 3), d((12), 4)) = \{11, 5\} = 11$$

Now, we cluster (12) & (34) to make (1234) with

$$d((12), (34)) = 11$$

(ii) In average linkage hierarchical procedure is taking $d((uv), w) = \frac{\sum_{i,j} d(x_i, y_j)}{n_u \times n_w}$

where x_i are elements in uv cluster & y_j are elements in w cluster.

$d(1,2)=1$ is the lowest observed distance.
 So we begin with clustering 1 & 2 to form (12).

$$d((12), 3) = \frac{d(1,3) + d(2,3)}{2} = \frac{1+2}{2} = 1.5$$

$$d((12), 4) = \frac{d(1,4) + d(2,4)}{2} = \frac{5+3}{2} = 4$$

So, the distance matrix is %

$$\begin{matrix} & (12) & 3 & 4 \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

$d(3,4)=3.5$ is the least distance observed here, so we combine 3 & 4 into a cluster (34).

$$d((12), (34)) = \frac{d((12), 3) + d((12), 4)}{2} = \frac{1.5+4}{2} = 2.75$$

~~So, we combine~~ The distance matrix now is %

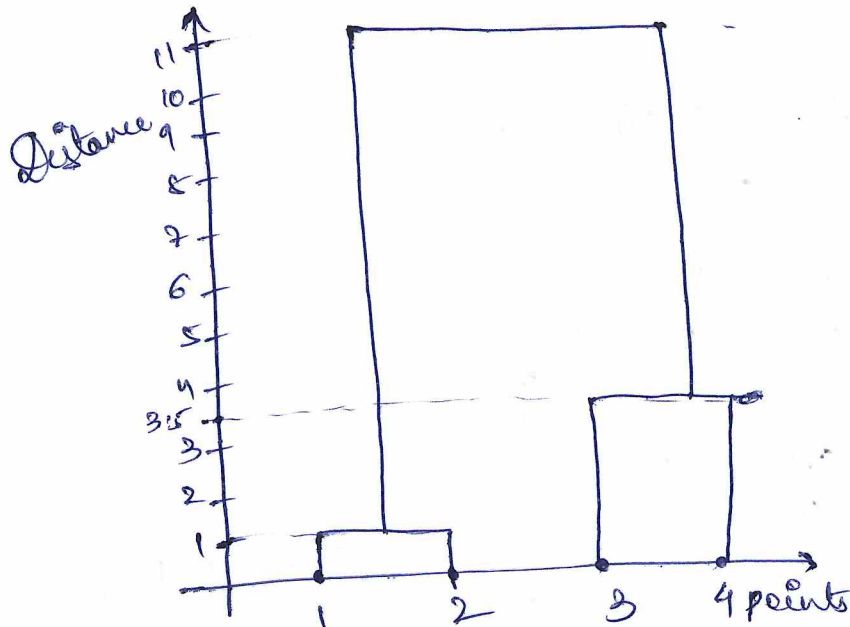
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$$\begin{matrix} & (12) & (34) \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 & \\ 2.75 & 0 \end{bmatrix} \end{matrix}$$

Now, we combine (12) & (34) to form 1 cluster (1234).

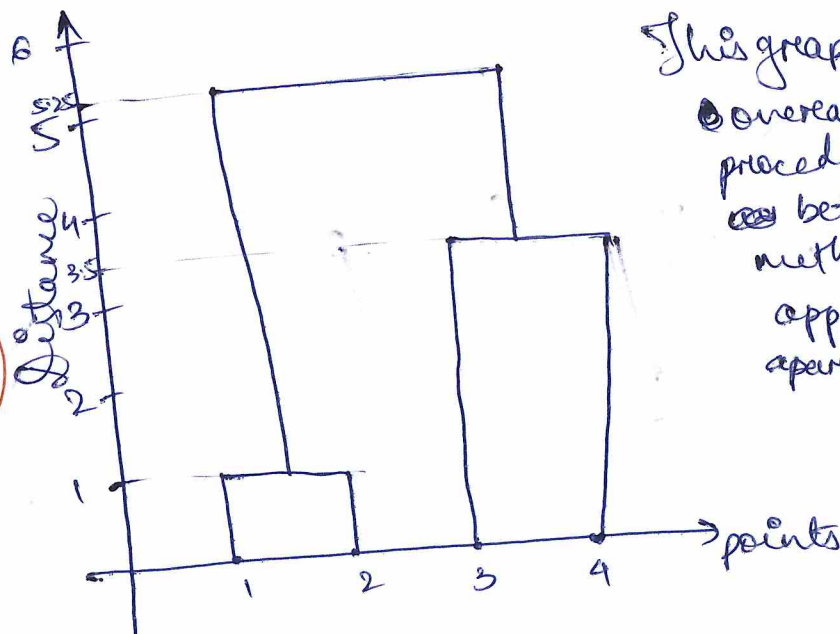
3. (b) Dendrograms :

for complete linkage problems :



This graph indicates that the overall clustering seemed very forceful & the clusters (1,2) & (3,4) appear to be very apart.

for average linkage procedure :



This graph indicates that overall clustering in this procedure was ~~not~~ ^{done} better than the previous method, (1,2) & (3,4) appear to be less far apart than (3,4,4).

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